# NINAD NAIK

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## EDUCATION

**Bachelor of Engineering in Information Technology** 2021 - 2025
University of Mumbai, Thadomal Shahani Engineering College (TSEC), Mumbai, India CGPA: **8.54 / 10**

## WORK EXPERIENCE

**Associate Backend Developer**, *Commotion* (acq. by Tata Communications) (gocommotion.com) Jan 2025 - Sep 2026

- Built and shipped **100+ enterprise integration connectors** in NestJS and TypeScript, exposing **3,200+ agent-callable actions** and **325 webhook triggers** for **Voice AI agents** across SAP, Oracle, Salesforce, ServiceNow and Workday.
- Cut **200 to 300 ms** from every agent tool call in the **Model Context Protocol (MCP)** gateway built with FastAPI by replacing a loopback HTTP call with an in-process handler, and added composite **PostgreSQL indexes** to its hot lookup path.
- Fixed an OAuth flow that broke under Kubernetes pod autoscaling by moving SSE session state to **Redis Pub/Sub**, and **load-tested** the MCP gateway at **40 to 500** concurrent voice-call sessions to size production pods.
- Implemented **OAuth 2.0** Auth Code and Client Credentials flows, extended **Agent-to-Agent (A2A)** agents to OAuth over **gRPC**, and remediated **Trivy SAST** findings across Docker images and Helm charts.
- Delivered integrations for **7 enterprise accounts** including Tata group companies and MNCs.

**Full Stack Developer Intern**, *Arxena Inc.* (arxena.com) Jun 2024 - Jul 2024

- Developed a **GPT-powered WhatsApp recruitment chatbot** in **NestJS, GraphQL and WebSockets** that screened candidates, stored resumes in **Amazon S3** and transcribed voice answers, with a **React.js** chat UI in **Twenty CRM**.
- Reduced a CRM server's **AWS EC2** cost by **75%** by running builds on on-demand instances launched from an **AMI** (details).

## OPEN SOURCE MENTORSHIP

**The Linux Foundation - LFX - Linux Kernel Mentorship Program mentee, Spring 2026** (link) Mar 2026 - Aug 2026

- Graduated with **11 patches** contributed to Linux kernel mainline; published a blog on the program.
- Converted text-based **Devicetree bindings** to JSON Schema; fixed documentation links, typos, and driver error handling.
- Used **cscope**, **Coccinelle** and **Smatch** for static analysis across **patch submission** and **review**; selected from **693 applicants**.

## PROJECTS

**Redspot**, *FastAPI, React.js, ClickHouse, Kafka, PostgreSQL, Docker* | (GitHub) Jun 2026

- Built an **event-driven website click analytics platform** in which a lightweight JavaScript tracker sends events to a FastAPI ingestion service, Kafka queues them and a consumer stores them in ClickHouse.
- Developed a React dashboard that renders target websites in an iframe and overlays a **color-coded click-density heatmap** drawn on canvas from live ClickHouse data, with desktop, tablet and mobile viewport previews.

**SpeakSure**, *Python, FastAPI, TensorFlow, React.js, MongoDB* | (Live app, GitHub) Oct 2025

- Engineered an **AI behavioural interview platform** that scores **spoken confidence** (1 to 5) with Wav2Vec2 and an MLP, with transcription, speech rate, filler words and **Gemini feedback**, analysed by a **background job queue on MongoDB**.
- Migrated inference from PyTorch and TensorFlow to **ONNX Runtime**, cutting about **3.5 GB** of ML dependencies from the server while running **19% faster** per chunk; models trained on a custom **751**-clip dataset (paper at ICCIS 2025).

**Shortomega**, *React.js, Next.js, Node.js, NestJS, TypeScript, Redis, Docker* | (Live app, GitHub) Nov 2024

- Designed a **full-stack URL shortener** on **Redis as the primary database** (Hashes, Sets, HyperLogLog, Lua scripts), with JWT auth, analytics and rate limiting.
- Dockerized and deployed the app publicly, and published a technical blog on using **Redis as a Primary Database**.

## SKILLS

**Languages**: TypeScript, JavaScript, Python, Go, C, C++, Java
**Backend & APIs**: NestJS, FastAPI, Node.js, REST, GraphQL, gRPC, OAuth 2.0, JWT
**AI**: MCP, A2A, RAG, LangChain, LangGraph, Agentic AI
**Web**: React.js, Next.js, Tailwind CSS, HTML, CSS, HeroUI
**Data & Messaging**: PostgreSQL, MySQL, MongoDB, Redis, Kafka, ClickHouse
**Platforms & Tools**: Linux, Docker, Kubernetes, AWS, CI/CD

## RESEARCH AND PUBLICATION

**Evaluating Human Speech Confidence with Deep Learning Techniques, Ninad Naik, et al.** Sep 2025

- Validated **Multilayer Perceptron (MLP)**, **Fuzzy Hypersphere Neural Network** and **Logistic Regression** models on **MFCC** and **Wav2Vec2** features of a **751-recording** dataset with 5-fold cross-validation, reaching **96% relaxed accuracy**.
- Presented at **ICCIS 2025**, BITS Pilani, Goa; accepted for publication in SCOPUS-indexed **Springer** LNNS. (Certificate)

## ADDITIONAL INFORMATION

- Open Source Contribution to the Linux Kernel, n8n.io and CircuitVerse.
- First Runner Up (CSS Battle 2022 & Javagenix 2022), 2nd Runner Up (PyExpo 2023), Top 10 (ACE Hacks 2024)
- Maintaining a **tech blog** to share experiences, learnings and results on tech.ninadnaik.me